M1 · EEE Module 1, electrical circuits, AC phasors, machines and safety

Electrical circuits, AC phasors, machines & safety

How to solve DC networks step-by-step (Mesh, Nodal, Superposition, Thevenin), the 4-way mega circuit, AC single/three-phase phasors, DC/Induction/BLDC motors, transformers, and earthing.

○ Mark as done ○ Highlight Quiz this module · FAT PYQs

All syllabus sections & lecture slide diagrams included · Ticks itself when every tab is read

Circuit Theorems ✓ 4-Way Mega Problem AC & 3-Phase Machines & BLDC Transformers Earthing & PYQs

Syllabus Order: 1. Components • 2. Mesh • 3. Nodal • 4. Superposition • 5. Thevenin • 6. Mega Circuit • 7. AC & Power • 8. Earthing • 9. Machines • 10. Drives
Sources: VIT Syllabus • VIT Mod 1 Notes • ExamCooker PYQs

M1T1 Module 1, topic 1

Electric Circuit Components, Ohm's Law, KCL, KVL & Divider Rules

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context	Before solving any circuit, picture every wire as a **water pipe** (VIT BAEEE101 Reference Material I). **Voltage (V, Volts)** is the water pressure pump; **Current (I = dq/dt, Amperes)** is the gallons/sec flow rate; and passive elements (R, L, C) react to that flow in three distinct ways:
Resistor (R, in Ω)	A narrow pinch in the pipe that resists current and turns electrical energy into heat ($P = I^2R = V^2/R$). Governed by **Ohm's Law: $V = I \cdot R$**.
Inductor (L, in Henrys H)	A heavy flywheel paddle inside the pipe: **$v(t) = L \cdot (di/dt)$**. It stores magnetic energy ($E_L = \frac{1}{2}LI^2$) and opposes sudden changes in **current**. In steady DC ($di/dt = 0$), an inductor acts as a **Short Circuit (plain wire, V = 0)**.
Capacitor (C, in Farads F)	A flexible rubber diaphragm across the pipe: **$i(t) = C \cdot (dv/dt)$**. It stores electric field energy ($E_C = \frac{1}{2}CV^2$) and opposes sudden changes in **voltage**. In steady DC ($dv/dt = 0$), a capacitor acts as an **Open Circuit (broken wire, I = 0)**.
Active vs. Passive & Bilateral vs. Unilateral	*Active* elements supply net energy (Voltage/Current sources, generators, Op-Amps); *Passive* elements absorb or store energy (R, L, C). R, L, and C are *Bilateral* (current flows equally well in either direction), whereas a PN diode is *Unilateral* (one-way).
Kirchhoff's Current Law (KCL — Node Law)	Based on **Conservation of Charge**. At any electrical junction (node), **$\Sigma I_{entering} = \Sigma I_{leaving}$**.
Kirchhoff's Voltage Law (KVL — Loop Law)	Based on **Conservation of Energy**. Around any closed loop, the algebraic sum of all voltage rises and drops is zero: **$\Sigma V = 0$**.

Voltage Division Rule (Series R1, R2)	Uses the *own* resistor on top: $V_1 = V_{total} \cdot [\, R_1 / (R_1 + R_2)\,]$.
Current Division Rule (Parallel R1, R2)	Current prefers the easier path, so put the *OTHER (opposite)* resistor on top: $I_1 = I_{total} \cdot [\, R_2 / (R_1 + R_2)\,]$ and $I_2 = I_{total} \cdot [\, R_1 / (R_1 + R_2)\,]$.

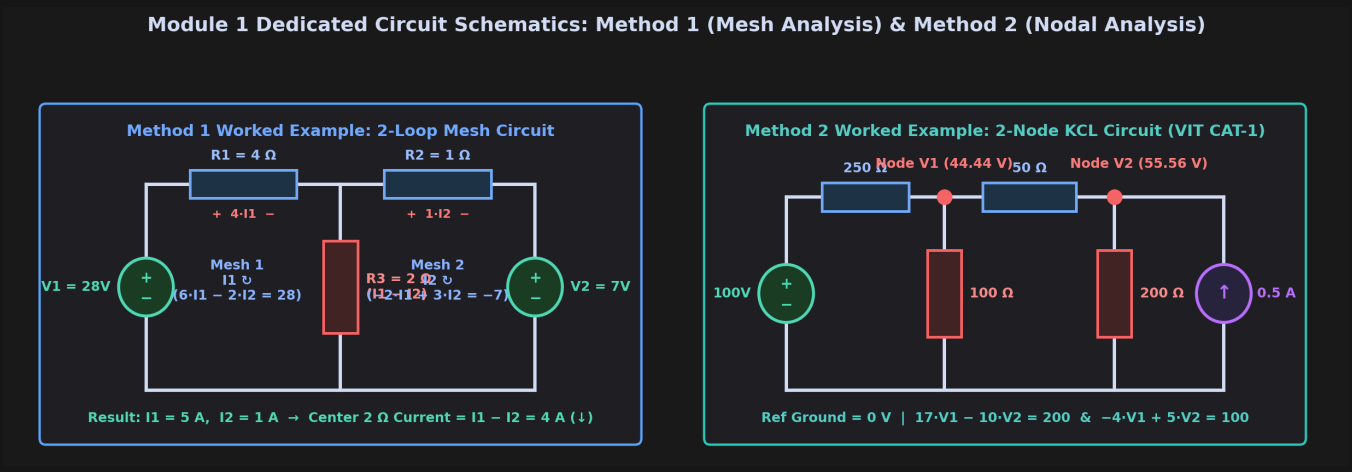


Figure 1.1: Dedicated circuit schematics for Topic 2 (Mesh Analysis Worked Example) and Topic 3 (Nodal Analysis Worked Example).

M1T2 Module 1, topic 2

Method 1 — Mesh Current Analysis (Step-by-Step Process & Best Example)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

When to use Mesh Analysis: Use Mesh Analysis whenever a planar circuit has many series resistors and voltage sources arranged in closed windows ("meshes"). Instead of solving for 5 or 6 separate branch currents, you only solve for 2 or 3 loop currents (VIT CAT-1 A1 Paper)!

Real-world intuition

The Foolproof 4-Step Process for Mesh Analysis (Never Lose a Sign!)

Draw Clockwise Loop Currents	Label every closed window with a clockwise loop current: I_1, I_2, I_3.
Identify Shared (Mutual) Branches	Any interior resistor sitting between Loop 1 and Loop 2 carries **both currents in opposite directions**. When you are writing the equation for Loop 1, the current through the shared resistor is $(I_1 - I_2)$. When you are writing the equation for Loop 2, it is $(I_2 - I_1)$!
Write KVL by Inspection (Matrix Form)	
For Mesh 1	For Mesh 1: **(Sum of all resistors in Mesh 1)** $\cdot I_1$ − **(Shared R_{12})** $\cdot I_2$ − **(Shared R_{13})** $\cdot I_3$ = Σ $V_{rises\ in\ Mesh\ 1}$·
Battery Sign Rule	Walk clockwise around your loop. If you enter the **(−) terminal and exit the (+) terminal** of a voltage

	source, it is a **Voltage Rise (+V on RHS)**. If you enter (+) and exit (−), it is a **Voltage Drop (−V on RHS)**.
Solve via Cramer's Rule	Write determinant **Δ** from the resistor matrix, replace column 1 with the voltage vector to get Δ_1, replace column 2 to get Δ_2, and compute $I_1 = \Delta_1 / \Delta$, $I_2 = \Delta_2 / \Delta$.

Dedicated Worked Example for Mesh Analysis (Refer to Figure 1.1 Left Circuit)

Overview & context	**Problem:** In the two-loop circuit of Figure 1.1 (Left), the left branch has a 28 V battery (+ on top) and R_1 = 4 Ω, the right branch has R_2 = 1 Ω and a 7 V battery (+ on top), and the shared center vertical branch has R_3 = 2 Ω. Find mesh currents I_1, I_2 and the downward current through the central 2 Ω resistor.
Step 1 (KVL in Mesh 1 — Clockwise I1)	Walking clockwise from bottom-left: we go from (−) to (+) through the 28 V battery (+28 V rise), drop across R_1 = 4 Ω, and drop across shared R_3 = 2 Ω with net current $(I_1 - I_2)$: $(4 + 2)I_1 - 2I_2 = 28 \Rightarrow \mathbf{6I_1 - 2I_2 = 28}$ — (Eq. 1)
Step 2 (KVL in Mesh 2 — Clockwise I2)	Walking clockwise in Mesh 2: we pass through shared R_3 = 2 Ω with net current $(I_2 - I_1)$, through R_2 = 1 Ω, and enter (+) to (−) of the 7 V battery (−7 V drop on RHS): $-2I_1 + (2 + 1)I_2 = -7 \Rightarrow \mathbf{-2I_1 + 3I_2 = -7}$ — (Eq. 2)
Step 3 (Cramer's Rule Solution)	System Determinant: $\mathbf{\Delta = (6)(3) - (-2)(-2) = 18 - 4 = 14}$. Numerator for I_1: $\mathbf{\Delta_1 = (28)(3) - (-2)(-7) = 84 - 14 = 70}$ ⇒ $\mathbf{I_1 = \Delta_1 / \Delta = 70 / 14 = 5.0\ A}$. Numerator for I_2: $\mathbf{\Delta_2 = (6)(-7) - (28)(-2) = -42 + 56 = 14}$ ⇒ $\mathbf{I_2 = \Delta_2 / \Delta = 14 / 14 = 1.0\ A}$.
Step 4 (Target Branch Current)	Downward current in the central 2 Ω resistor is $\mathbf{I_{2\Omega} = I_1 - I_2 = 5.0 - 1.0 = 4.0\ A\ (\downarrow)}$.

M1T3 Module 1, topic 3

Method 2 — Node Voltage Analysis (Step-by-Step Process & Best Example)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	**When to use Nodal Analysis:** Use Nodal Analysis whenever a circuit has parallel vertical branches, current sources, or many loops sharing just 1 or 2 top junctions (VIT CAT-1 B2 Paper). Even a 4-loop circuit often has only 2 principal nodes—meaning you only solve two equations!
Real-world intuition	The Foolproof 4-Step Process for Node Voltage Analysis
Pick Reference Ground (0 V) & Label Principal Nodes	Mark the bottom continuous wire as **0 V Reference Ground**. Label every top junction where 3 or more

	elements meet as unknown node voltages **V_1, V_2**.
The "Everyone Leaves My Node" Rule for Resistors	When standing at Node V_1, assume **every resistor current flows OUT of V_1**! By Ohm's law, current flowing out of V_1 toward another voltage V_{other} through resistor R is always: **$I_{branch} = (V_1 - V_{other}) / R$**.
Handle Current Sources Directly	Do NOT divide a current source by resistance—its current is already in Amperes! If a current source arrow points **INTO** your node, put **$+I_{source}$ on the RHS** (or $-I_{source}$ on the LHS). If it points **OUT** of your node, put **$-I_{source}$ on the RHS**.
Clear Fractions First	Multiply your KCL equation by the Least Common Multiple (LCM) of the denominators to get clean integer coefficients before solving!

Dedicated Worked Example for Nodal Analysis (Refer to Figure 1.1 Right Circuit — VIT CAT-1 B2)

Overview & context	**Problem:** In Figure 1.1 (Right), Node 1 (V_1) connects to a 100 V DC source via 250 Ω, to ground via 100 Ω, and to Node 2 (V_2) via 50 Ω. Node 2 connects to ground via 200 Ω and is fed by an upward 0.5 A current source. Find V_1, V_2, and the current flowing from Node 1 to Node 2 through the 50 Ω resistor.
Step 1 (KCL at Node V1 — all 3 resistor currents leave V1)	$(V_1 - 100) / 250 + (V_1 - 0) / 100 + (V_1 - V_2) / 50 = 0$. Multiply every term by LCM = 500: $2(V_1 - 100) + 5V_1 + 10(V_1 - V_2) = 0 \Rightarrow$ **$17V_1 - 10V_2 = 200$** — (Eq. 1)
Step 2 (KCL)	**Step 2 (KCL at Node V_2 — resistor currents leave V_2; 0.5 A source enters V_2):** $(V_2 - V_1) / 50 + (V_2 - 0) / 200 = +0.5$. Multiply every term by LCM = 200: $4(V_2 - V_1) + V_2 = 100 \Rightarrow$ **$-4V_1 + 5V_2 = 100$** — (Eq. 2)
Step 3 (Solve Simultaneously)	Multiply Eq. 2 by 2: $-8V_1 + 10V_2 = 200$. Add to Eq. 1: $9V_1 = 400 \Rightarrow$ **$V_1 = 44.44$ V**. Substitute V_1 into Eq. 2: $5V_2 = 100 + 4(44.44) = 277.78 \Rightarrow$ **$V_2 = 55.56$ V**.
Step 4 (Branch Current in 50 Ω from V1 to V2)	$I_{12} = (V_1 - V_2) / 50 = (44.44 - 55.56) / 50 =$ **-0.222 A** (meaning 0.222 A flows from right to left, from V_2 to V_1!).

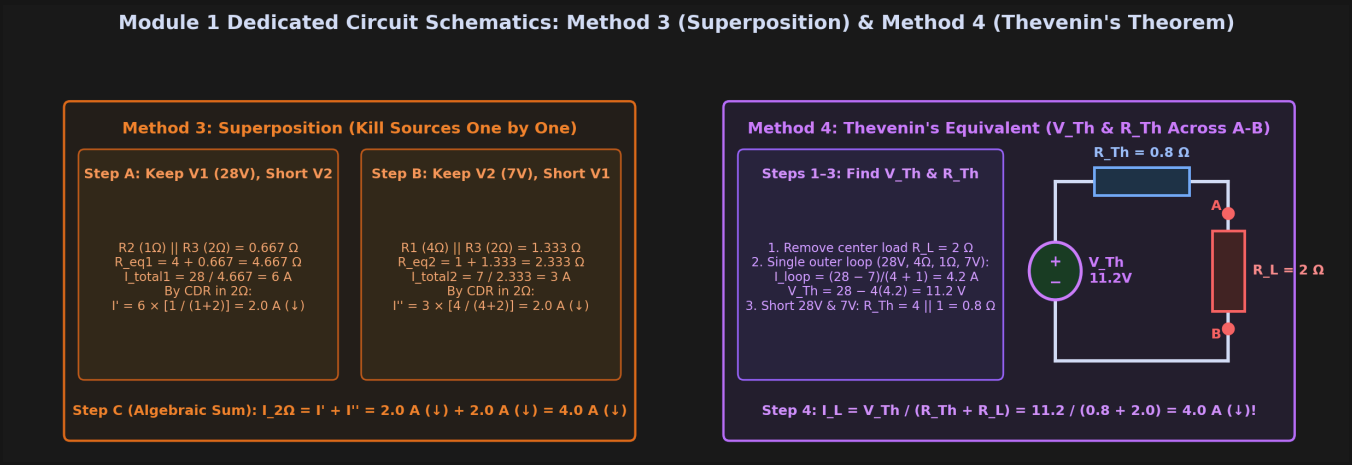


Figure 1.2: Step-by-step sub-circuits for Topic 4 (Superposition Theorem) and Topic 5 (Thevenin's Equivalent Circuit).

M1T4 Module 1, topic 4

Method 3 — Superposition Theorem (Step-by-Step Process & Best Example)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context

Statement of Superposition Theorem: In any linear, bilateral network containing two or more independent energy sources, the current through (or voltage across) any element equals the **algebraic sum** of the currents (or voltages) produced by each independent source acting **alone**, with all other independent sources deactivated (VIT Module 1 Reference Notes).

Real-world intuition

The Foolproof 4-Step Process for Superposition Theorem

Keep ONLY Source 1 Active; Kill All Other Sources	
To deactivate an ideal Voltage Source (V = 0)	To deactivate an ideal **Voltage Source (V = 0):** Replace it with a **Short Circuit (plain wire)**.
To deactivate an ideal Current Source (I = 0)	To deactivate an ideal **Current Source (I = 0):** Replace it with an **Open Circuit (erase the branch)**.
Find Partial Current I' (with Arrow Direction!)	Reduce series/parallel resistors to find total source current, then use the **Current Division Rule (CDR)** to find partial current **I'** in the target resistor. Always draw an arrow (↓ or ↑) next to I'!
Repeat for Source 2 Alone to Find I''	Reactivate Source 2, kill Source 1, and find partial current **I''** with its arrow direction.
Combine Algebraically	If I' and I'' point in the **SAME direction**, add them (**I = I' + I''**). If they point in **OPPOSITE directions**, subtract the smaller from the larger! *Warning:* Never add partial powers ($P' + P'' \neq P_{total}$ because $P = I^2R$ is non-linear; always find $P_{total} = I_{total}^2 \cdot R$ at the very end!).

Dedicated Worked Example for Superposition Theorem (Refer to Figure 1.2 Left)

Overview & context

Problem: Using the Superposition Theorem on the circuit of Figure 1.1 Left (V_1 = 28 V, R_1 = 4 Ω, central R_3 = 2 Ω, R_2 = 1 Ω, V_2 = 7 V), find the downward current in the central R_3 = 2 Ω resistor and total power dissipated in R_3.

Case A —

Case A — 28 V Source Active Alone (Short the 7 V battery with a wire, Figure 1.2 Left-A):

Right resistor R_2 (1 Ω) is now in parallel with central R_3 (2 Ω): R_{p1} = (1 × 2) / (1 + 2) = **2/3 Ω (0.667 Ω)**.

Total equivalent resistance seen by 28 V battery: $R_{eq1} = R_1 + R_{p1}$ = 4 + 2/3 = **14/3 Ω (4.667 Ω)**.

Total current leaving 28 V battery: I_{total1} = 28 / (14/3) = **6.0 A**.

By Current Division Rule (CDR), downward current in central 2 Ω resistor (put opposite 1 Ω on top!): **I' = 6.0 × [1 / (1 + 2)] = 2.0 A (↓)**.

Case B —

Case B — 7 V Source Active Alone (Short the 28 V battery with a wire, Figure 1.2 Left-B):

Left resistor R_1 (4 Ω) is now in parallel with central R_3 (2 Ω): R_{p2} = (4 × 2) / (4 + 2) = **4/3 Ω (1.333 Ω)**.

Total equivalent resistance seen by 7 V battery: $R_{eq2} = R_2 + R_{p2}$ = 1 + 4/3 = **7/3 Ω (2.333 Ω)**.

Total current leaving 7 V battery: I_{total2} = 7 / (7/3) = **3.0 A**.

By CDR, downward current in central 2 Ω resistor (put opposite 4 Ω on top!): **I'' = 3.0 × [4 / (4 + 2)] = 2.0 A (↓)**.

Case C — Algebraic Superposition & Power

Since both I' and I'' flow downward (↓), total current is $\mathbf{I_{2\Omega} = I' + I'' = 2.0 + 2.0 = 4.0\ A\ (\downarrow)}$. Power dissipated in 2 Ω resistor: $\mathbf{P_{2\Omega} = I_{2\Omega}^2 \cdot R_3 = (4.0)^2 \times 2 = 32.0\ W}$.

M1T5 Module 1, topic 5

Method 4 — Thevenin's Theorem & Max Power Transfer (Process & Best Example)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

Statement of Thevenin's Theorem: Any linear, bilateral two-terminal network containing voltage/current sources and resistors connected to a load resistor $\mathbf{R_L}$ across terminals A and B can be replaced by a single equivalent voltage source $\mathbf{V_{Th}}$ in series with a single equivalent resistance $\mathbf{R_{Th}}$ (VIT CAT-1 A1 Paper).

Real-world intuition

The Foolproof 4-Step Process for Thevenin's Theorem

Step 1 — Remove the Load Resistor RL

Pull out the target resistor R_L and mark its two open terminals as **A** and **B**.

Step 2 — Find Open-Circuit Voltage VTh (= VAB)

Calculate the voltage between open terminals A and B using simple KVL/KCL or Nodal analysis. *Exam Tip:* If a dangling resistor is connected in

	series with an open terminal A or B, **zero current flows through that dangling resistor**, so it has 0 V drop!
Step 3 — Find Thevenin Resistance RTh (= RAB)	Deactivate all independent sources (**Voltage Sources → Short Circuit; Current Sources → Open Circuit**) and calculate the equivalent resistance looking into terminals A and B.
Step 4 — Draw Thevenin Equivalent & Find IL	Connect V_{Th}, R_{Th}, and R_L in a single simple series loop: $\mathbf{I_L = V_{Th} / (R_{Th} + R_L)}$. (Bonus: Maximum power is transferred to the load when $\mathbf{R_L = R_{Th}}$, giving $\mathbf{P_{max} = V_{Th}^2 / (4 \cdot R_{Th})}$!).

Dedicated Worked Example for Thevenin's Theorem (Refer to Figure 1.2 Right)

Overview & context	**Problem:** Find the Thevenin equivalent circuit (V_{Th} and R_{Th}) across the central R_L = 2 Ω load resistor in Figure 1.1 Left (28 V with 4 Ω on left; 7 V with 1 Ω on right), hence find load current I_L and the maximum possible power that this network could deliver to an adjustable load.
Step 1 (Remove Load RL = 2 Ω)	Open terminals A (top node) and B (bottom ground wire).
Step 2 (Find VTh = VAB)	With the middle branch open, we just have ONE single outer loop containing the 28 V battery, 4 Ω, 1 Ω, and the opposing 7 V battery! Clockwise current around this outer loop is: $\mathbf{I_{outer} = (28 - 7) / (4 + 1) = 21 / 5 = 4.2\ A}$. Now walk from B (0 V) up the left branch to A: $\mathbf{V_{Th} = V_A = 28\ V - (4\ \Omega \times 4.2\ A) = 28 - 16.8 = 11.2\ V}$. (Check via right branch: 7 V + 1 Ω × 4.2 A = 11.2 V—exact match!).
Step 3 (Find RTh)	Short both the 28 V and 7 V batteries. Looking into terminals A and B, the 4 Ω left resistor and 1 Ω right resistor are directly in parallel: $\mathbf{R_{Th} = 4\ \Omega \,||\, 1\ \Omega = (4 \times 1) / (4 + 1) = 4 / 5 = 0.8\ \Omega}$.
Step 4 (Load Current & Max Power)	Reconnect R_L = 2 Ω in series with V_{Th} = 11.2 V and R_{Th} = 0.8 Ω (Figure 1.2 Right): $\mathbf{I_L = V_{Th} / (R_{Th} + R_L) = 11.2 / (0.8 + 2.0) = 11.2 / 2.8 = 4.0\ A\ (\downarrow)}$. If R_L were tuned for Maximum Power Transfer ($R_L = R_{Th}$ = 0.8 Ω): $\mathbf{P_{max} = V_{Th}^2 / (4R_{Th}) = (11.2)^2 / (4 \times 0.8) = 39.2\ W}$.

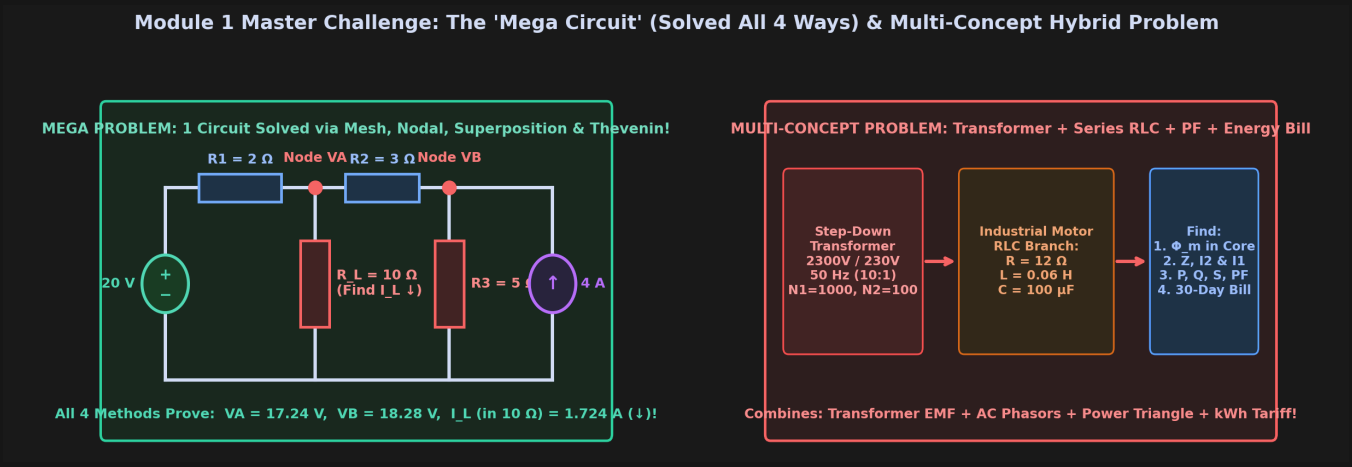


Figure 1.3: Left — The "Mega Circuit" solved separately by Mesh, Nodal, Superposition, and Thevenin. Right — Multi-Concept Hybrid Problem.

M1T6 Module 1, topic 6

The "Mega Circuit Challenge" — 1 Circuit Solved All 4 Ways Separately!

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

To make you 100% invincible in CAT-1 and FAT, let's take **one single tough mixed-source circuit** (Figure 1.3 Left: a **20 V voltage source** with series R_1 = 2 Ω on the left, our target vertical load resistor **R_L = 10 Ω** at Node A, a bridging resistor R_2 = 3 Ω to Node B, a vertical resistor R_3 = 5 Ω at Node B, and a **4 A upward current source** on the far right) and solve for the downward load current **I_L through the 10 Ω resistor** using **all 4 methods one by one**!

Mega Problem Solution 1: Using Node Voltage Analysis (Fastest for Mixed Sources!)

Label top-left junction

Label top-left junction above 10 Ω as **V_A**, top-right junction above 5 Ω as **V_B**, and bottom wire as **0 V Ground**.

KCL at Node VA

$(V_A - 20)/2 + V_A/10 + (V_A - V_B)/3 = 0$. Multiply by LCM = 30:

$15(V_A - 20) + 3V_A + 10(V_A - V_B) = 0 \Rightarrow$ **$28V_A - 10V_B = 300$** — (Eq. A)

KCL at Node VB (4 A enters Node B)

$(V_B - V_A)/3 + V_B/5 = 4$. Multiply by LCM = 15:

$5(V_B - V_A) + 3V_B = 60 \Rightarrow$ **$-5V_A + 8V_B = 60$** — (Eq. B)

Solve

From Eq. B, $V_A = (8V_B - 60)/5 = 1.6V_B - 12$. Substitute into Eq. A: $28(1.6V_B - 12) - 10V_B = 300 \Rightarrow 44.8V_B - 336 - 10V_B = 300 \Rightarrow 34.8V_B = 636 \Rightarrow$ **V_B = 18.276 V**, and **$V_A = 1.6(18.276) - 12 = 17.241$ V**.

Therefore, downward current in 10 Ω load is: **$I_L = V_A / 10 = 17.241 / 10 = 1.724$ A (↓)**.

Mega Problem Solution 2: Using Mesh Current Analysis

Assign 3 clockwise mesh currents

Assign 3 clockwise mesh currents: **I_1** (left window: 20V, 2Ω, 10Ω), **I_2** (middle window: 10Ω, 3Ω, 5Ω), and **I_3** (right window containing the 4 A upward current source on the right outer branch).

Mesh 3 Constraint (Free Gift!)	Because the right outer branch has a 4 A current source pointing UP (counter-clockwise), clockwise mesh current I_3 is immediately fixed: **$I_3 = -4.0$ A**!
KVL in Mesh 1	$(2 + 10)I_1 - 10I_2 = 20 \Rightarrow$ **$12I_1 - 10I_2 = 20$** — (Eq. 1)
KVL in Mesh 2	$-10I_1 + (10 + 3 + 5)I_2 - 5I_3 = 0$. Substitute $I_3 = -4$ A: $-10I_1 + 18I_2 - 5(-4) = 0 \Rightarrow$ **$-10I_1 + 18I_2 = -20$** — (Eq. 2)
Solve Eq. 1 & Eq. 2	$\Delta = (12)(18) - (-10)(-10) = 216 - 100 =$ **116**. $\Delta_1 = (20)(18) - (-10)(-20) = 360 - 200 = 160 \Rightarrow$ **$I_1 = 160 / 116 = 1.3793$ A**. $\Delta_2 = (12)(-20) - (20)(-10) = -240 + 200 = -40 \Rightarrow$ **$I_2 = -40 / 116 = -0.3448$ A**. Downward current in 10 Ω load is: **$I_L = I_1 - I_2 = 1.3793 - (-0.3448) = 1.724$ A (↓)** (Exact match!).
Mega Problem Solution 3: Using Superposition Theorem	
Case A (20 V Active Alone; Open-Circuit the 4 A Current Source)	With 4 A open, R_2 (3 Ω) and R_3 (5 Ω) are in series (3 + 5 = 8 Ω) in parallel with R_L (10 Ω): $R_p = (10 \times 8)/(10 + 8) = 80/18 = 4.444$ Ω. Total $R_{eq1} = 2 + 4.444 = 6.444$ Ω (= 116/18 Ω). Total current from 20 V: $I_{tot1} = 20 / (116/18) = 360/116 = 3.1034$ A. By CDR, downward current in 10 Ω branch: **$I' = 3.1034 \times [\ 8 / (10 + 8)\ ] = 160 / 116 = 1.3793$ A (↓)**.
Case B (4)	**Case B (4 A Current Source Active Alone; Short-Circuit the 20 V Battery):** Looking left from Node A, R_1 (2 Ω) $\|\|$ R_L (10 Ω) $= (2 \times 10)/(2 + 10) = 20/12 = 1.667$ Ω. That is in series with R_2 (3 Ω) = 4.667 Ω (= 14/3 Ω). Now the 4 A source splits between R_3 (5 Ω) and the left branch (14/3 Ω): current flowing left into Node A is $I_{left} = 4 \times [\ 5 / (5 + 14/3)\ ] = 20 / (29/3) = 60/29$ A. At Node A, this current splits between 2 Ω and 10 Ω; by CDR, downward current in 10 Ω is: **$I'' = (60/29) \times [\ 2 / (2 + 10)\ ] = 10 / 29 = 40 / 116 = 0.3448$ A (↓)**.
Total Superposition Current	**$I_L = I' + I'' = 1.3793 + 0.3448 = 1.724$ A (↓)** (Exact match!).
Mega Problem Solution 4: Using Thevenin's Theorem (Across Load R_L = 10 Ω)	
Step 1 (Remove RL = 10 Ω at Node A)	Find open-circuit voltage $V_{Th} = V_A$ and resistance R_{Th}.
Step 2 (Find RTh — Short 20 V, Open 4 A)	Looking into Node A, the left branch has 2 Ω to ground, and the right branch has (3 Ω + 5 Ω = 8 Ω) to ground. Thus: **$R_{Th} = 2\ \Omega\ \|\|\ 8\ \Omega = (2 \times 8) / (2 + 8) = 16 / 10 = 1.60\ \Omega$**!
Step 3 (Find VTh = VA(open))	Convert the 4 A current source in parallel with 5 Ω into an equivalent voltage source (4 A × 5 Ω = 20 V in series with 5 Ω), or write KCL at Node A with R_L removed: $(V_{Th} - 20)/2 + (V_{Th} - 20)/(3 + 5) = 0$! Since both ends are at 20 V, **$V_{Th} = 20.0$ V**!
Step 4 (Calculate IL in 5 Seconds)	**$I_L = V_{Th} / (R_{Th} + R_L) = 20.0 / (1.60 + 10.0) = 20.0 / 11.6 = 1.724$ A (↓)**! Look how all 4 methods give **1.724 A**—and see how each method shines!

M1T7 Module 1, topic 7

Single-Phase AC Circuits (RL, RC, RLC), Power Triangle & Power Factor

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

In AC circuits ($v(t) = V_m \sin(\omega t)$, where $V_{rms} = V_m / \sqrt{2} = 0.707\ V_m$, $V_{avg} = 2V_m / \pi = 0.637\ V_m$, Form Factor = 1.11, Peak Factor = 1.414), inductors and capacitors oppose current via frequency-dependent **Reactance (X)** (VIT Module 1 Notes).

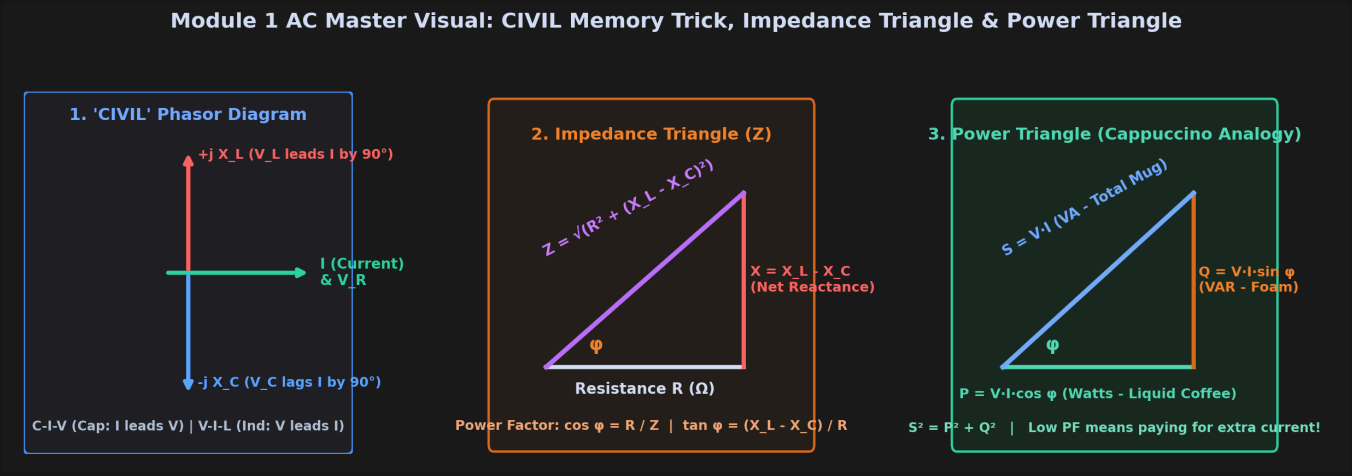


Figure 1.4: The CIVIL phase-angle rule, Impedance Triangle (R, X, Z), and Power Triangle (Active P, Reactive Q, Apparent S).

AC Circuit Topic	Impedance (Z) & Current (I)	Phase Angle (φ) & Power Factor	Power & Energy Equations
1. Series R-L Circuit	Inductive Reactance: $X_L = 2\pi fL$ $Z = R + jX_L = \sqrt{(R^2 + X_L^2)}$	Current I **lags** V by φ = $\tan^{-1}(X_L / R)$. **PF = cos φ = R / Z (Lagging)**.	**Active Power P = V · I · cos φ** (Watts) **Reactive Power Q = V · I · sin φ** (VAR)
2. Series R-C Circuit	Capacitive Reactance: $X_C = 1 / (2\pi fC)$ $Z = R - jX_C = \sqrt{(R^2 + X_C^2)}$	Current I **leads** V by φ = $\tan^{-1}(X_C / R)$. **PF = cos φ = R / Z (Leading)**.	**Apparent Power S = V · I = $\sqrt{(P^2 + Q^2)}$** (measured in VA or kVA).
3. Series R-L-C & Resonance	$Z = \sqrt{[R^2 + (X_L - X_C)^2]}$ **I = V / Z**	$\tan φ = (X_L - X_C) / R$. At **Resonance ($X_L = X_C$): $f_r = 1 / (2\pi\sqrt{(LC)})$**, Z = R, PF = 1.	**Electrical Energy (1 Unit = 1 kWh):** **E (kWh) = [P (Watts) × Hours] / 1000**. Bill = E (kWh) × Tariff Rate per Unit.

M1T8 Module 1, topic 8

Basics of Electrical Safety (Fuse, MCB, RCCB) & Earthing (Pipe vs. Plate)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

Why do we need both overcurrent breakers and earthing (VIT BAEEE101 Electrical Safety & Earthing Notes)? Because if a live wire touches the metal body of a refrigerator without earthing, anyone touching the fridge completes a path through their body (1,000 Ω wet resistance → 230 V / 1000 Ω = **230 mA > 50 mA**

fatal fibrillation threshold!). With **Earthing (< 2 to 5 Ω)**, a massive fault current rushes into the earth pit, instantly tripping the **MCB / RCCB**!

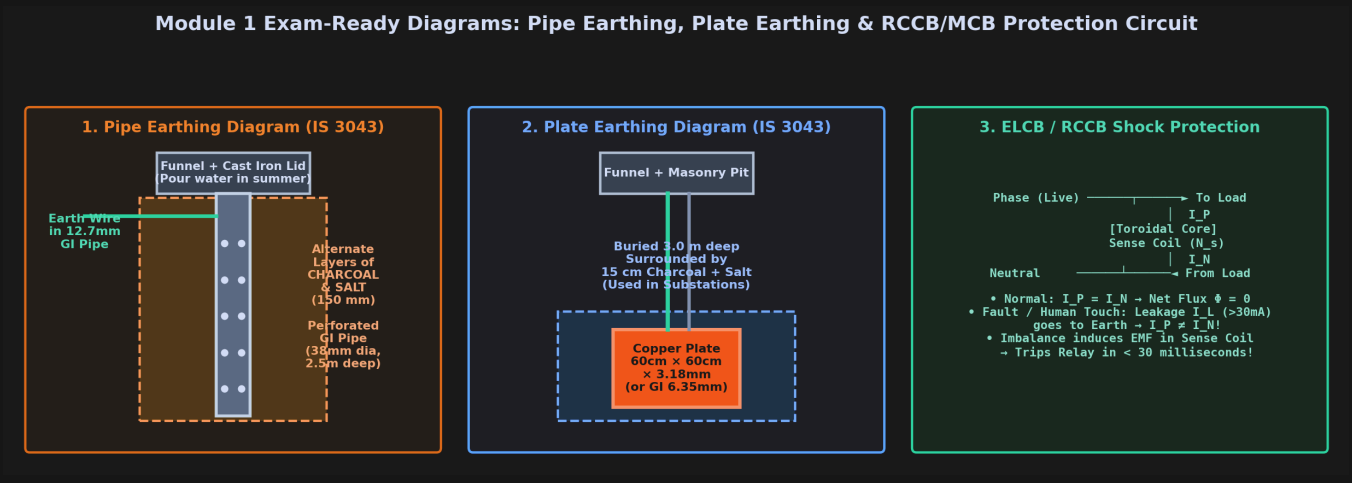


Figure 1.5: FAT Exam Diagrams for Pipe Earthing (IS 3043), Plate Earthing, and ELCB / RCCB Core-Balance Leakage Protection.

Safety / Earthing Topic	Working Principle & Construction	Fault Protected Against	Key Exam Numbers
Fuse vs. MCB vs. RCCB (ELCB)	• **Fuse:** Sacrificial low-melting Sn-Pb wire (melts via I^2Rt). • **MCB:** Resettable switch (Bimetallic strip for overload + Solenoid for short circuit). • **RCCB / ELCB:** Toroidal transformer detects $I_{Phase} \neq I_{Neutral}$.	• **Fuse & MCB:** Protect cables & appliances from Overload and Short-Circuit. • **RCCB / ELCB:** Protects **HUMANS from lethal electric shock** & earth leakage!	RCCB trips in **< 30 ms** when leakage current exceeds **30 mA**. Domestic wire colors: Red/Brown = Phase, Black/Blue = Neutral, **Green = Earth**.
Pipe Earthing vs. Plate Earthing (IS 3043)	• **Pipe Earthing (Figure 1.5-1):** Perforated GI pipe (**38 mm dia, 2.5 m deep**) buried vertically in alternate 150 mm layers of **Charcoal & Salt**. • **Plate Earthing (Figure 1.5-2):** Copper plate (**60×60 cm × 3.18 mm**) or GI plate (6.35 mm) buried **3 m deep**.	Keeps metal enclosures at **0 V Earth Potential** and provides a low-resistance fault path.	Why Charcoal & Salt? Retains soil moisture and boosts ionic conductivity so earth resistance stays **< 2 to 5 Ω** (tested by a **Megger Earth Tester**).

M1T9 Module 1, topic 9

Electrical Machines One-by-One (DC Motor, Induction Motor, BLDC, Transformer)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	Every electrical machine operates on either **Lorentz Force (F = B · I · L, Fleming's Left-Hand Rule for Motors)** or **Faraday's Law of Electromagnetic Induction (e = −N · dΦ/dt, Fleming's Right-Hand Rule for Generators & Mutual Induction for Transformers)** (VIT SELECT Course Plan).

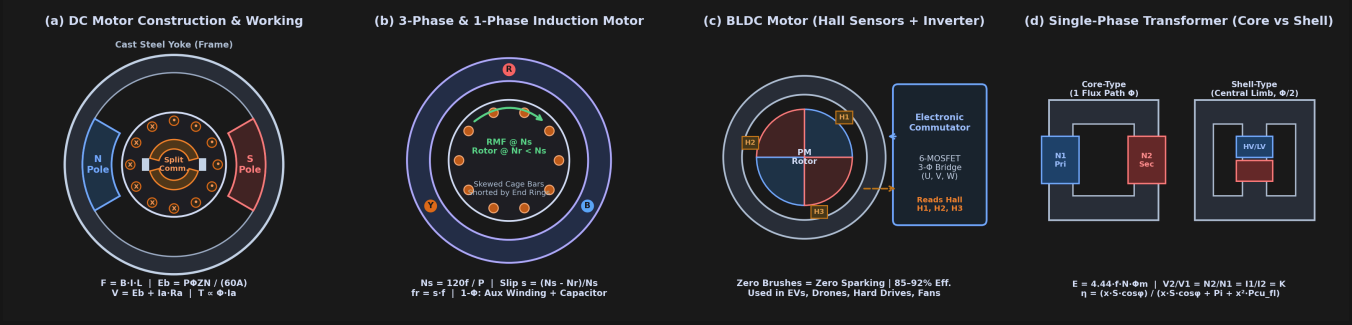


Figure 1.6: Detailed Construction & Working Diagrams — (a) DC Motor (Yoke, Poles, Split-Ring Commutator, Brushes), (b) 3-Phase & 1-Phase Induction Motor (RMF & Squirrel-Cage Rotor), (c) BLDC Motor (PM Rotor, Hall Sensors H1–H3 & 6-MOSFET Inverter Bridge), (d) Core-Type vs. Shell-Type Single-Phase Transformer.

Deep-Dive Machine Theory & Diagram Breakdown (High-Weightage FAT 10-Mark Reference)

DC Motor Construction & Working	**1. DC Motor Construction & Working:**
Stator (Field System)	Outer **Yoke** (cast steel frame providing mechanical protection and low-reluctance return path for half the pole flux Φ/2), laminated **Field Poles & Pole Shoes** (spread flux uniformly in the air gap), and **Field Winding** (carries field current I_f to establish main flux Φ).
Rotor (Armature System)	Slotted laminated silicon-steel **Armature Core** (reduces eddy-current and hysteresis losses), **Armature Winding** (Lap winding: A = P parallel paths for high-current, low-voltage traction; Wave winding: A = 2 parallel paths for high-voltage, low-current applications), **Split-Ring Copper Commutator** (insulated by 0.8 mm mica segments; acts as a mechanical inverter converting external DC into alternating current in the rotating coils so torque remains unidirectional!), and stationary spring-loaded **Carbon-Graphite Brushes** (self-lubricating; high contact resistance assists commutation).
Working & Governing Equations	By **Fleming's Left-Hand Rule**, a current-carrying conductor in flux density B experiences force **$F = B \cdot I \cdot L$**. As the armature rotates at N rpm, its conductors cut the magnetic flux and induce a counter-EMF (**Back EMF**) by Lenz's Law: **$E_b = (P \cdot \Phi \cdot Z \cdot N) / (60 \cdot A)$**. Voltage equation: **$V = E_b + I_a \cdot R_a$** (whereas for a DC Generator, $E_g = V + I_a \cdot R_a$). Gross mechanical power developed: **$P_m = E_b \cdot I_a$**, and Armature Torque: **$T_a = 9.55 \cdot (E_b \cdot I_a)/N = 0.159 \cdot \Phi \cdot Z \cdot I_a \cdot (P/A)$ N·m**. Speed relation: **$N \propto E_b / \Phi$** (reducing field flux Φ increases speed—hence a **DC Series Motor must NEVER be started at no-load** because $I_a \to 0$ implies $\Phi \to 0$ and $N \to \infty$ runaway!).
Three-Phase & Single-Phase Induction Motor (IM)	**2. Three-Phase & Single-Phase Induction Motor (IM):**
Rotating Magnetic Field (RMF)	When balanced 3-phase currents displaced by 120° flow through 3-phase stator windings displaced by 120°, they produce a resultant magnetic flux of constant magnitude **$\Phi_R = 1.5 \cdot \Phi_m$** revolving at **Synchronous Speed $N_s = 120 \cdot f / P$**.
Why Rotor Speed Nr Can Never Equal Ns	If $N_r = N_s$, relative speed between RMF and rotor bars is zero → zero flux cutting → induced rotor EMF $E_2 = 0$ → rotor current $I_2 = 0$ → electromagnetic torque T =

	0! Hence the rotor always slips behind N_s with fractional **Slip $s = (N_s - N_r)/N_s$** (typically 2%–5% at full load), and rotor current frequency is **$f_r = s \cdot f$**.
Squirrel-Cage vs. Slip-Ring (Wound) Rotor	*Squirrel-Cage* uses uninsulated copper/aluminium bars permanently short-circuited by end rings (skewed slightly to prevent magnetic locking/cogging and reduce magnetic hum; rugged, brush-free, moderate starting torque). *Slip-Ring* uses a 3-phase star-connected rotor winding brought out to 3 brass slip rings so external rheostat resistance can be inserted during starting (gives maximum starting torque at reduced starting current for cranes, hoists, and elevators, then shorted out at normal speed).
Why Single-Phase IM	A single alternating flux $\Phi = \Phi_m \cos(\omega t)$ resolves into two oppositely rotating fluxes of magnitude $\Phi_m/2$ ($+N_s$ forward and $-N_s$ backward). At standstill ($s = 1$), $T_{forward} = -T_{backward}$ → net starting torque = 0! Adding an **Auxiliary (Starting) Winding** in space quadrature (90°) in series with a **Capacitor** splits the single phase into two time-displaced currents (~80°–90° apart), creating a true revolving field and strong self-starting torque.
Brushless DC (BLDC) Motor & Electronic Commutation	**3. Brushless DC (BLDC) Motor & Electronic Commutation:**
Why BLDC Replaces Brushed DC	Mechanical carbon brushes and copper commutator segments in a conventional DC motor suffer from friction wear, carbon dust, acoustic noise, EMI/RF sparking, and speed limits. A BLDC motor turns the machine "inside-out": the **Permanent Magnet (NdFeB / SmCo) is placed on the Rotor**, and the **3-Phase Concentrated Armature Windings (U, V, W) are placed on the Stationary Stator** (allowing direct heat conduction through the outer housing).
Electronic Commutator Loop	Three **Hall-Effect Magnetic Position Sensors (H_1, H_2, H_3)** mounted 120° electrical apart on the stator detect the approaching N/S rotor poles and output a 3-bit digital code every 60° electrical rotation. A microcontroller reads (H_1, H_2, H_3) and fires the **6-MOSFET/IGBT 3-Phase Inverter Bridge** in a 6-step trapezoidal commutation sequence (two phases energized at any instant, one floating for Back-EMF sensing), keeping the stator magnetic field perpetually 90° ahead of the rotor for peak efficiency (85%–92%).
4. Single-Phase Transformer	
Working Principle	A static electromagnetic device that transfers AC electrical power from one circuit to another at the **exact same frequency f** via Faraday's Law of Mutual Induction. Alternating primary current I_0 sets up alternating flux $\Phi = \Phi_m \sin(\omega t)$ in the laminated 3% silicon-steel core, inducing RMS voltages **$E_1 = 4.44 \cdot f \cdot N_1 \cdot \Phi_m$** and **$E_2 = 4.44 \cdot f \cdot N_2 \cdot \Phi_m$**. Why a transformer burns out on DC: on DC ($f = 0$), inductive reactance $X_1 = 2\pi f L_1 = 0$ and $E_1 = 0$, so primary current $I_1 = V_{DC}/R_1$ becomes huge and roasts the winding insulation!
Core-Type vs. Shell-Type Construction	*Core-Type* has a single magnetic flux loop (Φ) with cylindrical concentric windings mounted on two outer vertical limbs (LV winding placed next to the grounded steel core to minimize insulation thickness, HV winding wound outside it; easier to dismantle for repair; preferred for high-voltage power transformers). *Shell-Type*

	has a 3-limb core where both HV and LV windings are interleaved ("sandwich/pancake coils") on the central limb, and the flux splits into two outer return paths (Φ/2 each); provides superior mechanical bracing against short-circuit forces.
Core (Iron) Losses vs. Copper Losses	*Iron Losses ($P_i = P_h + P_e$)* are constant at all loads (measured by Open-Circuit Test on LV side): **Hysteresis Loss $P_h = \eta_s \cdot B_{max}^{1.6} \cdot f \cdot V$** (minimized by using high-permeability cold-rolled grain-oriented CRGO 3% silicon steel) and **Eddy Current Loss $P_e = K_e \cdot B_{max}^2 \cdot f^2 \cdot t^2 \cdot V$** (minimized by stacking thin 0.35 mm–0.5 mm laminations coated with insulating varnish). *Copper Loss ($P_{cu} = I_1^2 R_1 + I_2^2 R_2 = x^2 \cdot P_{cu(FL)}$)* varies as the square of load fraction x (measured by Short-Circuit Test on HV side). Maximum efficiency occurs when **Variable Copper Loss = Constant Iron Loss ($x^2 \cdot P_{cu(FL)} = P_i \Rightarrow kVA_{\eta,max} = kVA_{FL} \cdot \sqrt{(P_i / P_{cu(FL)})}$)**.
Machine 1 — DC Motor	Converts DC electrical power into mechanical rotation. *Construction:* Stator (Cast-iron Yoke + Field Poles creating flux Φ), Rotor (Slotted Armature Core carrying armature conductors Z), **Split-Ring Commutator** (copper segments insulated by mica that act as a mechanical inverter—reversing armature coil current every 180° so torque stays unidirectional!), and Carbon Brushes. As the armature spins, it cuts flux and induces a **Back EMF: $E_b = (P \cdot \Phi \cdot Z \cdot N) / (60 \cdot A)$** (where A = 2 for Wave winding; A = P for Lap winding) that opposes supply V: **$V = E_b + I_a R_a$**.
Machine 2 — Induction Motor (3-Phase & 1-Phase)	The rugged workhorse of 90% of factories. When 3-phase AC feeds the stator windings, it creates a **Rotating Magnetic Field (RMF)** spinning at Synchronous Speed **$N_s = 120f / P$**. This RMF sweeps across the shorted rotor bars (**Squirrel-Cage Rotor** or **Slip-Ring Wound Rotor**), inducing current that makes the rotor chase the stator field at speed **$N_r < N_s$**. Why must N_r always be slightly less than N_s? Because if $N_r = N_s$, relative flux cutting becomes zero → induced current = 0 → torque = 0! Fractional difference is called **Slip: $s = (N_s - N_r) / N_s$** (typically 2%–5%). Note: A **1-Phase Induction Motor** (ceiling fan) produces a pulsating field and is *not self-starting*, so it uses an auxiliary winding + **Capacitor** to create a phase split at startup!
Machine 3 — BLDC Motor (Brushless DC Motor)	Turns the DC motor inside-out! **Permanent magnets** sit on the rotor, while the stator holds the coils. Instead of mechanical brushes and a sparking commutator, **Hall-Effect Rotor Position Sensors** tell an **Electronic Inverter Controller** which stator coils to energize next. Result: zero sparking, zero brush wear, silent operation, and **90%+ efficiency** (used in EVs, drones, CPU fans, hard drives).
Machine 4 — Single-Phase Transformer	A static device (zero moving parts!) that transfers AC power between two circuits at the **same frequency** via **Mutual Induction** on a laminated **CRGO Silicon Steel Core** (silicon reduces hysteresis loss; thin insulated laminations block eddy-current loss). **EMF Equation: $E_1 = 4.44 \cdot f \cdot N_1 \cdot \Phi_m$**; Voltage/Turns/Current ratio: **$V_2 / V_1 = E_2 / E_1 = N_2 / N_1 = I_1 / I_2 = K$**. *Exam Trap:* A transformer NEVER works on DC (in DC, f = 0 → $d\Phi/dt = 0$ → $E_1 = 0$ → primary winding draws huge current and burns out!).

M1T10 Module 1, topic 10

Power Electronics Converters & Industrial Electrical Drives

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	**Power Electronics** uses high-speed semiconductor switches (SCRs/Thyristors, IGBTs, Power MOSFETs) to convert electrical power from one form to another with minimal heat loss:
Rectifier (AC → DC Converter)	Converts fixed AC grid voltage into DC (used in phone/laptop chargers, EV battery fast-chargers, DC motor drives).
Inverter (DC → AC Converter)	Converts DC from batteries or solar panels into variable-voltage, variable-frequency AC (used in home UPS systems, rooftop Solar PV grid tie, and EV induction/BLDC traction drives).
Chopper (Fixed DC → Variable DC Converter)	High-speed electronic switch that acts as a "DC Transformer" (**Buck Chopper** steps DC down; **Boost Chopper** steps DC up—used in EV regenerative braking and battery management).
Cycloconverter / AC Voltage Controller (Fixed AC → Variable AC)	Directly changes AC voltage or steps down AC frequency (used in fan speed regulators, rolling mills, and giant cement kiln drives).
Overview & context	**Industrial Electrical Drive Architecture: [Electrical Power Source] → [Power Electronic Converter] → [Electric Motor] → [Mechanical Load (Pump/Robot/Spindle)]** controlled by a closed-loop **[Speed/Current Sensor + Microcontroller]**.

M1T11 Module 1, topic 11

Multi-Concept Combined Problem & Additional FAT PYQs

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	What if VIT combines **Transformers + Series RLC AC Phasors + Power Factor + Energy Tariff Calculation** into a single 10-mark FAT boss problem (Figure 1.3 Right)? Study this step-by-step solution and you will handle any hybrid problem with ease (ACM-VIT ExamCooker Archive)!

Multi-Concept Boss Problem [10 Marks]: Transformer + Series RLC Load + Power Triangle + 30-Day Bill

Overview & context	**Question (Refer to Figure 1.3 Right):** A 2300 V / 230 V, 50 Hz single-phase step-down transformer has N_1 = 1,000 primary turns and feeds an industrial motor-capacitor branch modeled as a series RLC circuit with R = 12 Ω, L = 0.06 H, and C = 100 μF across its 230 V secondary. Calculate: (a) Secondary turns N_2 and maximum core flux Φ_m, (b) Load reactances X_L, X_C and total impedance Z, (c) Secondary current I_2 and primary current I_1, (d) Load Power Factor cos φ and Active/Reactive/Apparent Power (P, Q, S), and (e) The 30-day electricity bill if this load runs 8 hours/day at ₹7.00 per unit (kWh).

Part (a) Transformer Parameters (2 Marks)	Turns ratio: $N_2 = N_1 \times (V_2 / V_1) = 1{,}000 \times (230 / 2300) =$ **100 turns**. From $E_2 = 4.44 \cdot f \cdot N_2 \cdot \Phi_m$: $230 = 4.44 \times 50 \times 100 \times \Phi_m \Rightarrow$ **$\Phi_m = 230 / 22{,}200 = 0.01036$ Wb (10.36 mWb)**.
Part (b) AC Reactances & Impedance (2 Marks)	Inductive reactance: **$X_L = 2\pi fL = 2 \times 3.1416 \times 50 \times 0.06 = 18.85\ \Omega$**. Capacitive reactance: **$X_C = 1 / (2\pi fC) = 1 / (2 \times 3.1416 \times 50 \times 100 \times 10^{-6}) = 31.83\ \Omega$**. Total Impedance: **$Z = \sqrt{[\ R^2 + (X_L - X_C)^2\ ]} = \sqrt{[\ 12^2 + (18.85 - 31.83)^2\ ]} = \sqrt{(144 + 168.48)} = 17.68\ \Omega$**.
Part (c) Secondary & Primary Currents (2 Marks)	Secondary load current: **$I_2 = V_2 / Z = 230 / 17.68 = 13.01$ A**. Primary high-voltage current: **$I_1 = I_2 \times (N_2 / N_1) = 13.01 \times (100 / 1{,}000) = 1.301$ A**.
Part (d) Power Factor & Power Triangle (2 Marks)	Power Factor: **$\cos\varphi = R / Z = 12 / 17.68 = 0.679$ Leading** (since $X_C > X_L$); $\sin\varphi = \lvert X_L - X_C \rvert / Z = 12.98 / 17.68 = 0.734$. Active Power: **$P = V_2 \cdot I_2 \cdot \cos\varphi = 230 \times 13.01 \times 0.679 = 2{,}031.8$ W (2.032 kW)**. Reactive Power: **$Q = V_2 \cdot I_2 \cdot \sin\varphi = 230 \times 13.01 \times 0.734 = 2{,}196.4$ VAR (2.196 kVAR)**. Apparent Power: **$S = V_2 \cdot I_2 = 230 \times 13.01 = 2{,}992.3$ VA (2.992 kVA)**.
Part (e) 30-Day Energy Consumption & Electricity Bill (2 Marks)	Total Energy in kWh (Units): **E = P (kW) × Hours/day × Days = 2.0318 kW × 8 h × 30 days = 487.63 kWh (Units)**. Total 30-Day Bill at ₹7.00/unit: **Cost = 487.63 × ₹7.00 = ₹3,413.41**.

Bonus Machine Numerical [5 Marks — VIT FAT]: 3-Phase Induction Motor Synchronous Speed & Slip

Overview & context	**Question:** A 3-phase, 4-pole, 50 Hz squirrel-cage induction motor runs at a full-load rotor speed of 1,440 RPM. Calculate: (i) Synchronous speed N_s, (ii) Percentage slip %s, and (iii) Rotor current frequency f_r.
(i) Synchronous Speed	**$N_s = (120 \cdot f) / P = (120 \times 50) / 4 = 1{,}500$ RPM**.
(ii) Percentage Slip	**$s = (N_s - N_r) / N_s = (1500 - 1440) / 1500 = 60 / 1500 = 0.04$ (or 4.0% Slip)**.
(iii) Rotor Frequency	**$f_r = s \cdot f = 0.04 \times 50$ Hz $= 2.0$ Hz**.